The Differences Between Direct Alignment Algorithms are a Blur

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RLHF vs DAA

- RLHF = SFT + reward modeling + reward maximisation
- **SFT**: $\mathscr{L}_{\text{SFT}}\left(\pi_\theta, x, y\right) = -\log \pi_\theta(y \mid x)$ (per sample, per dataset – expectation)
- **reward modeling**:
$$\mathscr{L}_{\text{RM}}\left(r_\psi\right) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathscr{D}}\left[\log \sigma\left(r_\psi\left(x, y_w\right) - r_\psi\left(x, y_l\right)\right)\right]$$
- **reward maximisation**: generating responses that maximise the learned reward + using KL penalty to prevent reward hacking
- $\max_{\pi_\theta} \mathbb{E}_{x\sim\mathscr{D}, y\sim\pi_\theta(y|x)}\left[r_\phi(x, y)\right] - \beta \mathbb{D}_{\text{KL}}\left[\pi_\theta(x, y) \| \pi_{\text{ref}}(x, y)\right]$

RLHF vs DAA

- DAA = Direct Alignment Algorithm (SFT + Alignment Step)
- no reward modelling, no reward maximisation (= no RL)
- alignment step: using preference-optimization loss functions

- $r_\theta^{\text{ref}}(y, x) = \log\left(\frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}\right)$ from DPO (Rafailov et al., 2023), which acts as an implicit reward $\beta\, r_\theta^{\text{ref}}$. No length normalization is used.
- $r_\theta^{\text{odds}}(y, x) = \log\left(\frac{\pi_\theta(y|x)}{1-\pi_\theta(y|x)}\right)$ utilized in ORPO (Hong et al., 2024), representing the odds of generating y versus not generating it. While not directly derived from an RL objective in the same way as r_θ^{ref}, its empirical success in methods like ORPO and ASFT motivates its inclusion in our comparative analysis.

$$\mathscr{L}_{\text{DPO}} = -\log\sigma\left(\beta r_\theta^{\text{ref}}\left(y_w, x\right) - \beta r_\theta^{\text{ref}}\left(y_l, x\right)\right)$$ – literally reward modelling

RLHF vs DAA

- single-stage alignment: SFT and alignment steps are merged into one
- combined loss:

$$\mathscr{L}_{\text{Single}}\left(\pi_\theta\right) = \mathbb{E}_{(x,y_w,y_l)\sim\mathscr{D}}\left[\mathscr{L}_{\text{SFT}}\left(\pi_\theta, x, y_w\right) + \lambda\mathscr{L}_{\text{Align}}\left(\pi_\theta, x, y_w, y_l\right)\right]$$

Odds Ratio Preference Optimization (ORPO) (Hong et al., 2024) is defined as: $\mathcal{L}_{\text{ORPO}} = -\log \pi_\theta(y_w|x) - \lambda \underbrace{\log \sigma\left(r_\theta^{\text{odds}}(y_w, x) - r_\theta^{\text{odds}}(y_l, x)\right)}_{-\mathcal{L}_{\text{ORPO}_{\text{Align}}}}$. pairwise

Aligned Supervised Fine-Tuning (ASFT) (Wang et al., 2024) is defined as: $\mathcal{L}_{\text{ASFT}} = -\log \pi_\theta(y_w|x) - \lambda\Big(\underbrace{\log \sigma\left(r_\theta^{\text{odds}}(y_w, x)\right) + \log \sigma\left(-r_\theta^{\text{odds}}(y_l, x)\right)}_{-\mathcal{L}_{\text{ASFT}_{\text{Align}}}}\Big)$. pointwise

On the Difference Between Direct Alignment Algorithms

- r_θ^{ref} vs. r_θ^{odds}
- r_θ^{ref} originates in RLHF
- r_θ^{odds} is derived from odds-based reasoning
- $r_\theta^{\text{ref}}(y, x) = \log\left(\frac{\pi_\theta(y \mid x)}{\pi_{\text{ref}}(y \mid x)}\right)$ ~KL-divergence – (DPO, implicit reward)
- $r_\theta^{\text{odds}}(y, x) = \log\left(\frac{\pi_\theta(y \mid x)}{1 - \pi_\theta(y \mid x)}\right)$ – (ORPO, odds of generating y versus not generating)

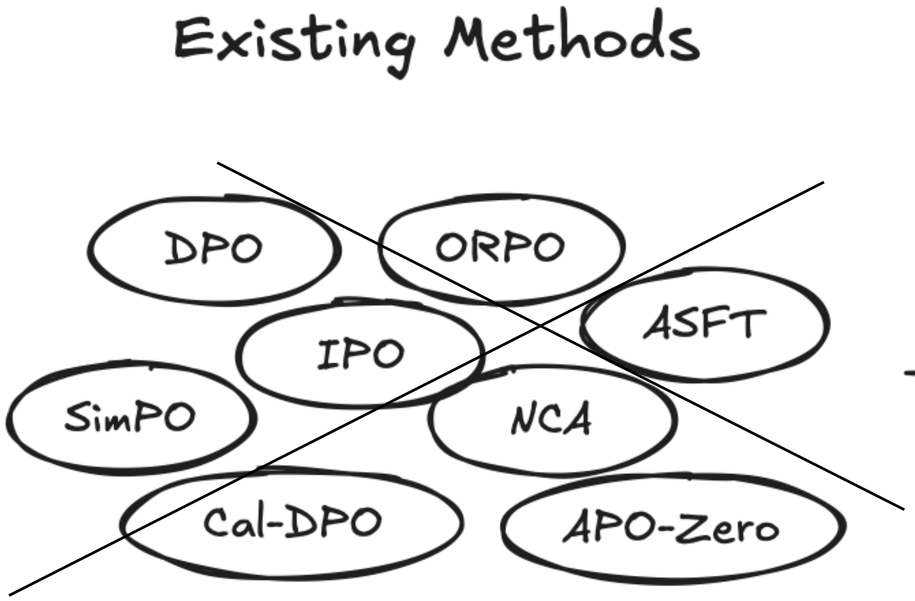
- pairwise methods (DPO, IPO, SimPO, ORPO) depend on **relative reward differences** between candidate texts
- pointwise methods (APO-Zero, NCA, Cal-DPO, ASFT) maximize the probability of chosen sequences and minimize that of rejected ones **independently** of their mutual gap
- spoiler: pairwise objectives often yield more robust ranking signals than pointwise ones

Alignment Algorithm	Loss Function
Direct Preference Optimization (DPO)	$\mathcal{L}_{\text{DPO}} = -\log\sigma\left(\beta r_\theta^{\text{ref}}(y_w, x) - \beta r_\theta^{\text{ref}}(y_l, x)\right)$
Identity Preference Optimization (IPO)	$\mathcal{L}_{\text{IPO}} = \left(r_\theta^{\text{ref}}(y_w, x) - r_\theta^{\text{ref}}(y_l, x) - \frac{1}{2\beta}\right)^2$
Simple Preference Optimization (SimPO)	$\mathcal{L}_{\text{SimPO}} = -\log\sigma\left(\beta\log\pi_\theta(y_w, x) - \beta\log\pi_\theta(y_l, x) - \gamma\right)$
Noise Contrastive Alignment (NCA)	$\mathcal{L}_{\text{NCA}} = -\log\sigma\left(\beta r_\theta^{\text{ref}}(y_w, x)\right) - 0.5\log\sigma\left(-\beta r_\theta^{\text{ref}}(y_w, x)\right) - 0.5\log\sigma\left(-\beta r_\theta^{\text{ref}}(y_l, x)\right)$
Calibrated Direct Preference Optimization (Cal-DPO)	$\mathcal{L}_{\text{Cal-DPO}} = -\log\sigma\left(r_\theta^{\text{ref}}(y_w, x) - r_\theta^{\text{ref}}(y_l, x)\right) + \left(r_\theta^{\text{ref}}(y_w, x) - \frac{1}{2\beta}\right)^2 + \left(r_\theta^{\text{ref}}(y_l, x) + \frac{1}{2\beta}\right)^2$
Anchored Preference Optimization Zero (APO-Zero)	$\mathcal{L}_{\text{APO-Zero}} = -\sigma\left(\beta r_\theta^{\text{ref}}(y_w, x) + \sigma\left(\beta r_\theta^{\text{ref}}(y_l, x)\right)\right)$

- **DPO:** $f(r_w - r_l)$ – pairwise + beta
- **IPO**: $\log\sigma(\ldots) \rightarrow$ another $f(\ldots), f(r_w - r_l)$ – pairwise + beta
- **SimPO**: $f(r_w - r_l)$ – pairwise + beta
- **CalDPO**: calibrated (DPO + IPO), $f(r_w - r_l) + f(r_w) + f(r_l)$ – mixed + beta
- **APO-Zero**: $f(r_w) + f(r_l)$ – pointwise + beta
- **ORPO, ASFT**: no SFT => no reference policy; $r_\theta^{ref} \rightarrow r_\theta^{odds}$ **(!)**, no beta **(!)**

At first glance, it may seem unnecessary to introduce β into single-stage methods, yet we will demonstrate that neither the single-stage design nor the absence of β is mandatory for ASFT and ORPO

- **RQ1**: does an **explicit SFT stage** improve the alignment quality of ORPO and ASFT?
- **RQ2**: does the **tempering factor** enhance the alignment quality of ASFT and ORPO?
- **RQ3**: which **factors** of DAAs **affect alignment quality**?



1. Unification

Method	SFT	β
DPO	✓	✓
IPO	✓	✓
SimPO	✓	✓
APO-Zero	✓	✓
NCA	✓	✓
Cal-DPO	✓	✓
ORPO	X → ✓	X → ✓
ASFT	X → ✓	X → ✓

Unification Part 1: SFT

3.1.1 ORPO and ASFT can operate without the SFT loss term and as two-stage methods

First, note that $\mathcal{L}_{\text{ASFT}_{\text{Align}}} = -\log \pi_\theta(y_w|x) - \log\big(1 - \pi_\theta(y_l|x)\big)$, and thus $\mathcal{L}_{\text{ASFT}} = -(1 + \lambda) \log \pi_\theta(y_w|x) - \lambda \log\big(1 - \pi_\theta(y_l|x)\big)$; see Appendix B for a proof. Second, $\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{ASFT}} + \lambda \log\big(\pi_\theta(y_w|x)(1 - \pi_\theta(y_l|x)) + \pi_\theta(y_l|x)(1 - \pi_\theta(y_w|x))\big)$; see Appendix C for details.

From these equations it follows that $\mathcal{L}_{\text{ORPO}} \leq \mathcal{L}_{\text{ASFT}}$ and $\mathcal{L}_{\text{ORPO}_{\text{Align}}} \leq \mathcal{L}_{\text{ASFT}_{\text{Align}}}$ (see Appendix C.2).

These results lead to three observations: (i) $\mathcal{L}_{\text{ASFT}}$ upper-bounds $\mathcal{L}_{\text{ORPO}}$; therefore, minimizing the former automatically minimizes the latter. (ii) $\mathcal{L}_{\text{ASFT}_{\text{Align}}}$ can be regarded as the simplest DAA loss, mirroring the structure of BCE (see Appendix B.3); (iii) Most importantly, the SFT term in these losses is already encapsulated by the alignment term, implying that explicit inclusion of the SFT loss may be redundant. Thus, we hypothesize that the SFT term can be omitted, motivating our **RQ1**: *"Does an explicit SFT stage improve the alignment quality of ORPO and ASFT?"* and experiments in Section 5.1, where we compare ASFT and ORPO both in their original single-stage form and in a two-stage variant that follows an explicit SFT phase.

Unification Part 2: β

While $\mathcal{L}_{\text{ASFT}_{\text{Align}}}$ and $\mathcal{L}_{\text{ORPO}_{\text{Align}}}$ use r_θ^{odds}, many DAAs incorporate a scaling parameter β. To enable a unified comparison and investigate the role of β, we introduce it to scale r_θ^{odds}:

$$\mathcal{L}^{\beta}_{\text{ASFT}_{\text{Align}}} = -\log \sigma(\beta r_\theta^{\text{odds}}(y_w, x)) - \log \sigma(-\beta r_\theta^{\text{odds}}(y_l, x)), \tag{1}$$

$$\mathcal{L}^{\beta}_{\text{ORPO}_{\text{Align}}} = -\log \sigma(\beta r_\theta^{\text{odds}}(y_w, x) - \beta r_\theta^{\text{odds}}(y_l, x)). \tag{2}$$

Both $\mathcal{L}^{\beta}_{\text{ASFT}}$ and $\mathcal{L}^{\beta}_{\text{ORPO}}$ generalize their vanilla counterparts (recovering them when $\beta = 1$). As in DPO, β can be viewed as a *temperature* or *scaling* parameter that regulates the intensity of the preference for "good" odds. See Appendix Section D for for gradient formulations and more details on these methods.

Results

RQ1: does an explicit SFT stage improve the alignment quality of ORPO and ASFT?

Init	Method	LC% (std)	WR% (std)	AH% (CI)
Base	SFT	6.7 (0.43)	4.5 (0.63)	3.5 (-0.7, 0.8)
SFT	ORPO	**24.1** (0.84)	17.8 (1.17)	15.3 (-1.6, 1.8)
SFT	ASFT	16.4 (0.72)	11.9 (0.99)	10.6 (-1.2, 1.3)
Base	ORPO†	14.8 (0.71)	10.3 (0.95)	8.4 (-1.3, 1.3)
Base	ASFT†	14.5 (0.73)	10.2 (0.94)	7.5 (-1.1, 1.2)
SFT	ORPO†	13.4 (0.69)	9.3 (0.91)	7.7 (-0.9, 1.1)
SFT	ASFT†	11.4 (0.63)	7.5 (0.83)	7.5 (-1.1, 1.1)
SFT	DPO	23.4 (0.85)	**20.0** (1.18)	**17.5** (-1.8, 1.8)

Table 1: **Base and SFT-initialized alignment methods on the Llama 3.1 8B model with the UF dataset.** SFT-initialized methods demonstrate better performance compared to their traditional formulations without $\mathcal{L}_{\text{SFT}}$. Results marked with † correspond to training with $\mathcal{L}_{\text{SFT}}$, using the best hyperparameters: $\text{lr} = 1 \times 10^{-6}$ for ORPO and $\text{lr} = 7 \times 10^{-7}$ for ASFT. For other setups, the best hyperparameters are: $\text{lr} = 5 \times 10^{-7}$ for standard SFT ORPO/ASFT, and $\text{lr} = 1 \times 10^{-5}/6 \times 10^{-6}$ for Base ORPO/ASFT.

- TLDR: **yes**
- LC – length-controlled win rate, WR – win rates, AH – Arena Hard
- ORPO achieves results comparable to classical DPO in both LC Win Rate and AH WR metrics
- ASFT shows notable gains in AH WR after the SFT stage, although it still underperforms compared to ORPO or DPO
- $\mathscr{L}_{\text{ORPO}} \leq \mathscr{L}_{\text{ASFT}}$ and $\mathscr{L}_{\text{ORPO}_{\text{Align}}} \leq \mathscr{L}_{\text{ASFT}_{\text{Align}}}$

Results

RQ2: does the tempering factor (β) enhance the alignment quality of ASFT and ORPO?

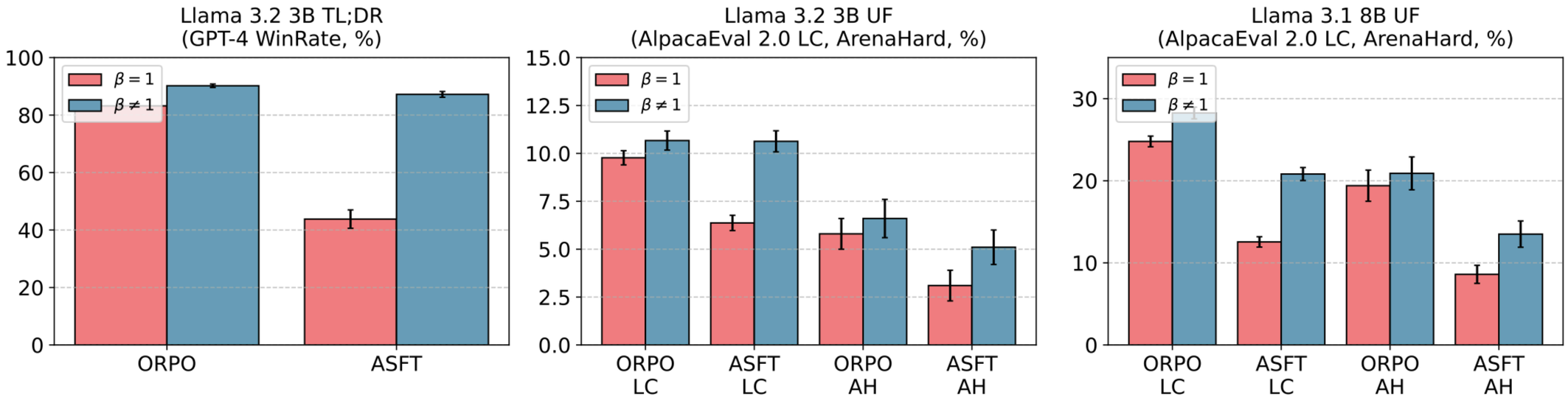


Figure 2: **Impact of the β Parameter on ASFT and ORPO Alignment Quality.** The plot shows how tuning β (Section 3.1.2) affects both ASFT and ORPO performance. Results are reported for GPT-4 Win Rate in the Llama 3.2 3B TL;DR setup and for AlpacaEval 2 LC Win Rate in the Llama 3.1 8B UF scenario. All other hyperparameters (e.g., learning rates) are selected via grid search, using each method's best configuration at $\beta = 1$ as the baseline. See Section 5.2 for more details.

Results

RQ3: which factors of DAAs affect alignment quality?

	Method	Llama 3.2 3B UF			Llama 3.1 8B UF		
		AlpacaEval 2		ArenaHard	AlpacaEval 2		ArenaHard
		LC% (std)	WR% (std)	WR% (CI)	LC% (std)	WR% (std)	WR% (CI)
	SFT	5.02 (0.34)	3.21 (0.55)	1.4 (-0.4, 0.4)	10.27 (0.54)	5.44 (0.70)	2.6 (-0.5, 0.6)
pairwise	DPO	**11.43** (0.58)	11.79 (0.99)	6.8 (-1.0, 0.9)	26.82 (0.77)	23.69 (1.25)	19.0 (-1.9, 1.8)
pairwise	IPO	11.24 (0.60)	11.67 (1.01)	**6.8** (-1.0, 1.1)	28.18 (0.83)	24.43 (1.26)	19.1 (-1.6, 1.5)
pairwise	SimPO	10.56 (0.44)	11.94 (0.95)	6.4 (-1.0, 1.1)	27.65 (0.77)	25.62 (1.29)	**21.5** (-1.9, 1.9)
pairwise	ORPO	10.67 (0.50)	**12.23** (0.97)	6.6 (-1.0, 1.1)	**28.25** (0.71)	**28.59** (1.33)	20.9 (-2.0, 2.0)
pointwise	APO Zero	10.36 (0.53)	11.22 (0.98)	6.0 (-1.0, 0.9)	23.15 (0.76)	19.03 (1.18)	17.3 (-1.8, 1.8)
pointwise	NCA	10.33 (0.53)	11.02 (0.97)	5.1 (-0.7, 0.8)	23.21 (0.80)	18.67 (1.17)	15.1 (-1.5, 1.6)
pointwise	Cal-DPO	10.62 (0.57)	10.15 (0.94)	4.8 (-0.9, 0.9)	23.19 (0.82)	18.85 (1.18)	15.2 (-1.5, 1.6)
pointwise	ASFT	10.63 (0.55)	9.21 (0.88)	5.1 (-0.9, 0.9)	20.82 (0.79)	16.34 (1.13)	13.5 (-1.6, 1.5)

Table 2: **AlpacaEval 2 and ArenaHard Results for Llama 3.2 3B and Llama 3.1 8B UF.** The SFT model was trained on the UltraChat dataset. The best hyperparameters for each method were selected according to Section 4.2. Bold values indicate the best performance for each benchmark, while underlined values represent the second-best performance. See Section 5.3 for more details.

Results

RQ3: which factors of DAAs affect alignment quality?

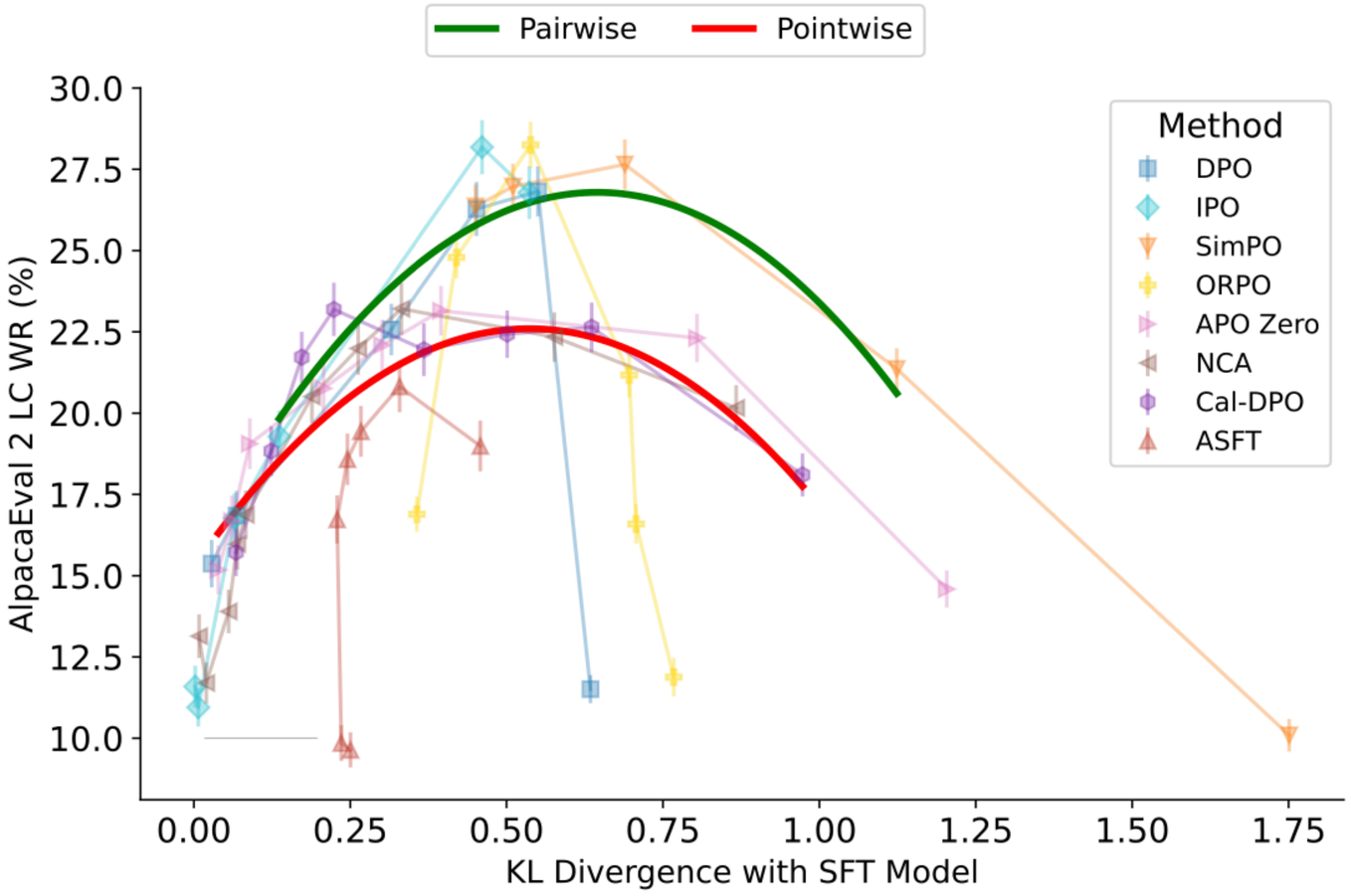


Table 3: **Pareto front for alignment quality and KL divergence.** Results for Llama 3.1 8B UF on AlpacaEval 2 LC. Methods are grouped into pairwise and pointwise categories, with pairwise achieving higher LC values while remaining within overlapping confidence intervals. See Section 5.3 for more details.

Results

RQ3: which factors of DAAs affect alignment quality?

1. Unification

Method	SFT	β
DPO	✓	✓
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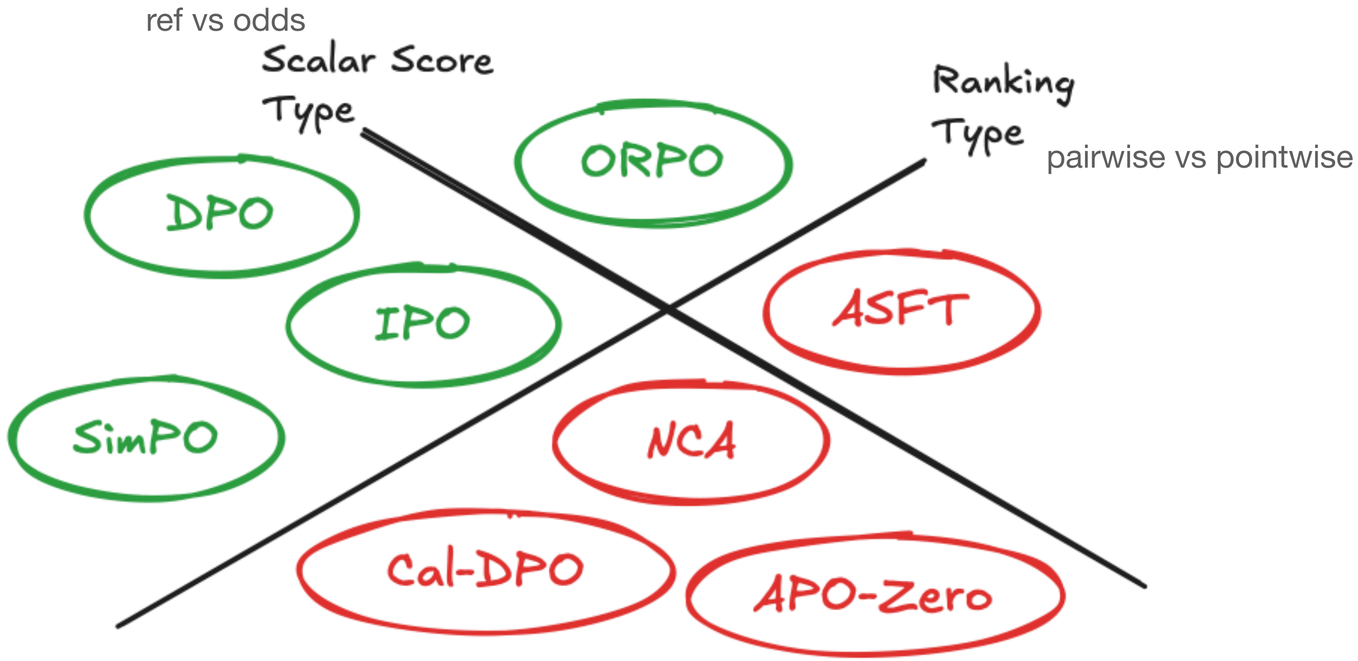
SimPO	✓	✓
APO-Zero	✓	✓
NCA	✓	✓
Cal-DPO	✓	✓
ORPO	X → ✓	X → ✓
ASFT	X → ✓	X → ✓

2. Quality Factors



Discussion

Why do pairwise objectives outperform pointwise ones?

- both r_θ^{ref} and r_θ^{odds} rely on a shared scoring function $r_\theta(x, y)$, with bias reflected in the marginalised score $\mathbb{E}_y[r_\theta(x, y)]$
- models can either (i) improve rankings on complex examples or (ii) refine existing rankings (e.g., widening y_w vs. y_l gaps)
- pointwise methods drive $p_\theta(y_w \mid x) \to 1$ and $p_\theta(y_l \mid x) \to 0$, pushing $\mathbb{E}_y[r_\theta(x, y)] \to 0$, which consumes capacity and hinders learning harder tasks
- pairwise training only enforces y_w scores above y_l (controlled by β), preserving $\mathbb{E}_y[r_\theta(x, y)]$ and avoiding capacity waste
- performance regimes:
  - hard tasks: both methods perform similarly (insufficient capacity for unlearning)
  - easy tasks: pointwise methods catch up (excess capacity)
  - Intermediate tasks: pointwise suffers due to misaligned capacity (unlearning bias but not harder examples)

Conclusion

We provide a unified analysis of Direct Alignment Algorithms (DAAs), showing that the main axis of difference is not the scalar score (r^{ref} or r^{odds}), but the choice between pairwise and pointwise objectives. Pairwise methods outperform pointwise, particularly at the 8B scale, as they preserve the marginalized preference structure in the data, whereas pointwise methods tend to distort it by shifting prompt-specific biases. Introducing an explicit SFT stage and parameter β benefits odds-ratio based DAAs (we show its applicability to ORPO and ASFT), with most gains realized using just 5-10% of the SFT data. This finding clarifies why previous claims of "best" DAA Meng et al. (2024); Xiao et al. (2024); Wang et al. (2024) often depend on underexplored details of setup and bias.